# Formulario: Electrodinámica Avanzada

## ONDAS PLANAS Y POLARIZACIÓN

$$\mathbf{E}(\mathbf{r},t) = \mathrm{Re}\left[\mathbf{E}_0 e^{i(\mathbf{k}\cdot\mathbf{r}-\omega t)}\right], \quad \mathbf{B}(\mathbf{r},t) = \mathrm{Re}\left[\mathbf{B}_0 e^{i(\mathbf{k}\cdot\mathbf{r}-\omega t)}\right]$$

$$\nabla^2\mathbf{E} - \mu\epsilon\frac{\partial^2\mathbf{E}}{\partial t^2} = 0, \quad \nabla^2\mathbf{H} - \mu\epsilon\frac{\partial^2\mathbf{H}}{\partial t^2} = 0$$

$$v = \frac{1}{\sqrt{\mu\epsilon}}, \quad n = \frac{c}{v} = \sqrt{\frac{\epsilon\mu}{\epsilon_0\mu_0}}, \quad Z = \sqrt{\frac{\mu}{\epsilon}} = \frac{Z_0}{n}$$

$$\mathbf{k}\cdot\mathbf{E}_0 = 0, \quad \mathbf{k}\cdot\mathbf{B}_0 = 0, \quad k^2 = \mu\epsilon\omega^2$$

$$\mathbf{B}_0 = \frac{1}{\omega}\mathbf{k}\times\mathbf{E}_0, \quad \mathbf{H}_0 = \frac{1}{Z}\hat{\mathbf{k}}\times\mathbf{E}_0$$

$$\langle u\rangle = \frac{1}{4}\epsilon|\mathbf{E}_0|^2 + \frac{1}{4\mu}|\mathbf{B}_0|^2 = \frac{1}{2}\epsilon|\mathbf{E}_0|^2$$

$$\mathbf{S} = \mathbf{E}\times\mathbf{H}, \quad \langle\mathbf{S}\rangle = \frac{1}{2}\mathrm{Re}(\mathbf{E}_0\times\mathbf{H}_0^*) = \frac{|\mathbf{E}_0|^2}{2Z}\hat{\mathbf{k}}$$

$$\mathbf{J} = \begin{pmatrix} E_x \\ E_y \end{pmatrix} = \begin{pmatrix} a_x e^{i\delta_x} \\ a_y e^{i\delta_y} \end{pmatrix}, \quad \Delta = \delta_y - \delta_x$$

$$\frac{E_x^2}{a_x^2} + \frac{E_y^2}{a_y^2} - 2\frac{E_xE_y}{a_xa_y}\cos\Delta = \sin^2\Delta$$

$$S_0 = a_x^2 + a_y^2, \quad S_1 = a_x^2 - a_y^2$$

$$S_2 = 2a_xa_y\cos\Delta, \quad S_3 = 2a_xa_y\sin\Delta$$

$$S_0^2 \geq S_1^2 + S_2^2 + S_3^2 \quad \text{(Esfera de Poincaré)}$$

## INTERFACES Y COEFICIENTES DE FRESNEL

$$\theta_r = \theta_i, \quad n_1\sin\theta_i = n_2\sin\theta_t \quad \text{(Snell)}$$

$$r_s = \frac{n_1\cos\theta_i - n_2\cos\theta_t}{n_1\cos\theta_i + n_2\cos\theta_t}, \quad t_s = \frac{2n_1\cos\theta_i}{n_1\cos\theta_i + n_2\cos\theta_t}$$

$$r_p = \frac{n_2\cos\theta_i - n_1\cos\theta_t}{n_2\cos\theta_i + n_1\cos\theta_t}, \quad t_p = \frac{2n_1\cos\theta_i}{n_2\cos\theta_i + n_1\cos\theta_t}$$

$$R_{s,p} = |r_{s,p}|^2, \quad T_{s,p} = \frac{n_2\cos\theta_t}{n_1\cos\theta_i}|t_{s,p}|^2, \quad R + T = 1$$

$$\tan\theta_B = \frac{n_2}{n_1} \implies r_p = 0 \quad \text{(Ángulo de Brewster)}$$

$$\theta_i > \theta_c = \arcsin\left(\frac{n_2}{n_1}\right), \quad \kappa = \frac{\omega}{c}\sqrt{n_1^2\sin^2\theta_i - n_2^2}$$

## MEDIOS DISPERSIVOS, CONDUCTORES Y PLASMAS

$$\nabla^2\mathbf{E} - \mu\epsilon\frac{\partial^2\mathbf{E}}{\partial t^2} - \mu\sigma\frac{\partial\mathbf{E}}{\partial t} = 0 \quad (\mathbf{J}_f = \sigma\mathbf{E})$$

$$\epsilon_c(\omega) = \epsilon + i\frac{\sigma}{\omega}, \quad k_c^2 = \mu\epsilon_c(\omega)\omega^2, \quad k_c = \beta + i\frac{\alpha}{2}$$

$$\sigma \gg \omega\epsilon \implies \beta \approx \frac{\alpha}{2} \approx \sqrt{\frac{\mu\sigma\omega}{2}}, \quad \delta = \frac{2}{\alpha} = \sqrt{\frac{2}{\mu\sigma\omega}}$$

$$\omega_p = \sqrt{\frac{Ne^2}{m\epsilon_0}}, \quad \epsilon(\omega) = \epsilon_0\left(1 - \frac{\omega_p^2}{\omega^2}\right), \quad \omega^2 = \omega_p^2 + c^2k^2$$

$$v_\phi = \frac{\omega}{k} = \frac{c}{\sqrt{1 - \omega_p^2/\omega^2}}, \quad v_g = \frac{d\omega}{dk} = c\sqrt{1 - \frac{\omega_p^2}{\omega^2}}$$

$$v_\phi v_g = c^2, \quad k(\omega) \approx k(\omega_0) + \frac{1}{v_g}(\omega - \omega_0) + \frac{1}{2}k''(\omega_0)(\omega - \omega_0)^2$$

## CAUSALIDAD Y KRAMERS-KRONIG

$$\mathbf{P}(t) = \epsilon_0\int_{-\infty}^{t}\chi(t-t')\mathbf{E}(t')dt' \implies \chi(-\omega) = \chi^*(\omega)$$

$$\mathrm{Re}\,\chi(\omega) = \frac{1}{\pi}\mathcal{P}\int_{-\infty}^{\infty}\frac{\mathrm{Im}\,\chi(\omega')}{\omega' - \omega}d\omega'$$

$$\mathrm{Im}\,\chi(\omega) = -\frac{1}{\pi}\mathcal{P}\int_{-\infty}^{\infty}\frac{\mathrm{Re}\,\chi(\omega')}{\omega' - \omega}d\omega'$$

$$\mathrm{Re}\,\epsilon(\omega) - 1 = \frac{2}{\pi}\mathcal{P}\int_0^{\infty}\frac{\omega'\mathrm{Im}\,\epsilon(\omega')}{\omega'^2 - \omega^2}d\omega'$$

$$\mathrm{Im}\,\epsilon(\omega) = -\frac{2\omega}{\pi}\mathcal{P}\int_0^{\infty}\frac{\mathrm{Re}\,\epsilon(\omega') - 1}{\omega'^2 - \omega^2}d\omega'$$

$$\epsilon(\omega) = 1 + \int_0^{\infty}\frac{f(\Omega)}{\Omega^2 - \omega^2 - i\Gamma\omega}d\Omega$$

$$\lim_{\Gamma\to 0^+}\mathrm{Im}\,\epsilon(\omega) = \frac{\pi}{2\omega}f(\omega) \quad (\omega > 0)$$

## GUÍAS DE ONDA

$$\mathbf{E} = [\mathbf{E}_\perp + E_z\hat{\mathbf{z}}]\,e^{i(kz-\omega t)}, \quad \mathbf{B} = [\mathbf{B}_\perp + B_z\hat{\mathbf{z}}]\,e^{i(kz-\omega t)}$$

$$(\nabla_\perp^2 + \gamma^2)E_z = 0, \quad (\nabla_\perp^2 + \gamma^2)B_z = 0, \quad \gamma^2 = \frac{\omega^2}{c^2} - k^2$$

$$\mathbf{E}_\perp = \frac{i}{\gamma^2}\left(k\nabla_\perp E_z - \omega\hat{\mathbf{z}}\times\nabla_\perp B_z\right)$$

$$\mathbf{B}_\perp = \frac{i}{\gamma^2}\left(k\nabla_\perp B_z + \frac{\omega}{c^2}\hat{\mathbf{z}}\times\nabla_\perp E_z\right)$$

$$\text{TM: } B_z = 0,\ E_z|_S = 0; \quad \text{TE: } E_z = 0,\ \left.\frac{\partial B_z}{\partial n}\right|_S = 0$$

$$\text{Guía Rec: } \omega_{mn} = c\gamma_{mn} = c\sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2}$$

$$k_{mn}(\omega) = \frac{1}{c}\sqrt{\omega^2 - \omega_{mn}^2}, \quad v_\phi v_g = c^2$$

$$Z_{\mathrm{TE}} = \frac{Z_0}{\sqrt{1 - (\omega_{mn}/\omega)^2}}, \quad Z_{\mathrm{TM}} = Z_0\sqrt{1 - \left(\frac{\omega_{mn}}{\omega}\right)^2}$$

$$\alpha = \frac{P_{\mathrm{dis}}}{2P_{\mathrm{trans}}}, \quad P_{\mathrm{dis}} = \frac{1}{2\sigma\delta}\oint_S|\mathbf{H}_{\mathrm{tang}}|^2dl$$

## POTENCIALES RETARDADOS Y RADIACIÓN

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi}\int\mathbf{J}(\mathbf{r}')\frac{e^{ik|\mathbf{r}-\mathbf{r}'|}}{|\mathbf{r}-\mathbf{r}'|}d^3r'$$

$$r \gg \lambda \implies \mathbf{A}(\mathbf{r}) \approx \frac{\mu_0}{4\pi}\frac{e^{ikr}}{r}\int\mathbf{J}(\mathbf{r}')e^{-ik\hat{\mathbf{r}}\cdot\mathbf{r}'}d^3r'$$

$$\mathbf{B} \approx ik\hat{\mathbf{r}}\times\mathbf{A}, \quad \mathbf{E} \approx c\mathbf{B}\times\hat{\mathbf{r}}$$

$$\mathbf{p} = \int\mathbf{r}'\rho(\mathbf{r}')d^3r', \quad \mathbf{A}(\mathbf{r}) = -\frac{i\mu_0\omega}{4\pi}\mathbf{p}\frac{e^{ikr}}{r}$$

$$\mathbf{B} = \frac{\mu_0\omega^2}{4\pi c}(\hat{\mathbf{r}}\times\mathbf{p})\frac{e^{ikr}}{r}, \quad \mathbf{E} = \frac{\mu_0\omega^2}{4\pi}\left[(\hat{\mathbf{r}}\times\mathbf{p})\times\hat{\mathbf{r}}\right]\frac{e^{ikr}}{r}$$

$$\frac{dP}{d\Omega} = \frac{\mu_0\omega^4}{32\pi^2c}|\mathbf{p}|^2\sin^2\theta, \quad P_{\mathrm{total}} = \frac{\mu_0\omega^4}{12\pi c}|\mathbf{p}|^2$$

$$R_{\mathrm{rad}} = \frac{2P_{\mathrm{total}}}{I_0^2} = \frac{\mu_0\omega^2d^2}{6\pi c} \quad \text{(Dipolo eléctrico } E1\text{)}$$

### ELECTRODINÁMICA COVARIANTE

$$\eta_{\mu\nu} = \eta^{\mu\nu} = \text{diag}(1, -1, -1, -1)$$

$$x^\mu = (ct, \mathbf{r}), \quad \partial_\mu = \left(\frac{1}{c}\frac{\partial}{\partial t}, \nabla\right), \quad \Box = \partial_\mu \partial^\mu = \frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \nabla^2$$

$$U^\mu = \gamma(c, \mathbf{v}), \quad p^\mu = mU^\mu = (E/c, \mathbf{p}), \quad p_\mu p^\mu = m^2c^2$$

$$J^\mu = (c\rho, \mathbf{J}), \quad A^\mu = (\Phi/c, \mathbf{A}), \quad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$$

$$F^{\mu\nu} = \begin{pmatrix} 0 & -E_x/c & -E_y/c & -E_z/c \\ E_x/c & 0 & -B_z & B_y \\ E_y/c & B_z & 0 & -B_x \\ E_z/c & -B_y & B_x & 0 \end{pmatrix}$$

$$\tilde{F}^{\mu\nu} = \frac{1}{2}\epsilon^{\mu\nu\rho\sigma}F_{\rho\sigma}, \quad (\mathbf{E}/c \to \mathbf{B},\ \mathbf{B} \to -\mathbf{E}/c)$$

$$\partial_\mu F^{\mu\nu} = \mu_0 J^\nu, \quad \partial_\mu \tilde{F}^{\mu\nu} = 0, \quad \partial_\mu J^\mu = 0$$

$$\partial_\mu A^\mu = 0 \implies \Box A^\mu = \mu_0 J^\mu \quad \text{(Gauge de Lorenz)}$$

$$F_{\mu\nu}F^{\mu\nu} = 2\left(B^2 - \frac{E^2}{c^2}\right), \quad F_{\mu\nu}\tilde{F}^{\mu\nu} = -\frac{4}{c}(\mathbf{E}\cdot\mathbf{B})$$

### DINÁMICA RELATIVISTA Y DRIFTS

$$E'_x = E_x, \quad B'_x = B_x$$
$$E'_y = \gamma(E_y - vB_z), \quad B'_y = \gamma\left(B_y + \frac{v}{c^2}E_z\right)$$
$$E'_z = \gamma(E_z + vB_y), \quad B'_z = \gamma\left(B_z - \frac{v}{c^2}E_y\right)$$

$$K^\mu = \frac{dp^\mu}{d\tau} = qF^{\mu\nu}U_\nu \implies \frac{d\mathbf{p}}{dt} = q(\mathbf{E} + \mathbf{v}\times\mathbf{B})$$

$$L = -mc^2\sqrt{1 - \frac{v^2}{c^2}} - q\Phi + q\mathbf{A}\cdot\mathbf{v}$$

$$\mathbf{P}_{\text{can}} = \mathbf{p} + q\mathbf{A}, \quad H = \sqrt{(\mathbf{P} - q\mathbf{A})^2c^2 + m^2c^4} + q\Phi$$

$$\mathbf{v}_E = \frac{\mathbf{E}\times\mathbf{B}}{B^2}, \quad \mu = \frac{mv_\perp^2}{2B} = \text{const} \implies \frac{B_{\text{esp}}}{B_0} = \frac{v_0^2}{v_{\perp 0}^2}$$

### ACCIÓN Y TENSORES DE ENERGÍA

$$\mathcal{L}_{\text{Maxwell}} = -\frac{1}{4\mu_0}F_{\mu\nu}F^{\mu\nu} - J_\mu A^\mu$$

$$\mathcal{L}_{\text{Proca}} = -\frac{1}{4\mu_0}F_{\mu\nu}F^{\mu\nu} + \frac{\mu_\gamma^2}{2\mu_0}A_\mu A^\mu \quad \text{(Masa fotón)}$$

$$T^{\mu\nu}_{\text{can}} = -\frac{1}{4\mu_0}\eta^{\mu\nu}F_{\alpha\beta}F^{\alpha\beta} - \frac{1}{\mu_0}F^{\mu\rho}\partial^\nu A_\rho$$

$$\Theta^{\mu\nu} = T^{\mu\nu}_{\text{can}} + \frac{1}{\mu_0}F^{\mu\rho}\partial_\rho A^\nu = \frac{1}{\mu_0}\left(F^{\mu\rho}F^\nu{}_\rho + \frac{1}{4}\eta^{\mu\nu}F_{\alpha\beta}F^{\alpha\beta}\right)$$

## 1. Guías de Onda Rectangulares ($0 \le x \le a$, $0 \le y \le b$)

Ecuación de Helmholtz: $(\nabla_\perp^2 + \gamma^2)\psi = 0$, con $\gamma^2 = \frac{\omega^2}{c^2} - k^2$.

1. **Modo TM:** $B_z = 0$, hallar $E_z$. Condición de Dirichlet: $E_z|_{\partial\Omega} = 0$.

$$E_z(x, y) = E_0 \sin\left(\frac{m\pi x}{a}\right)\sin\left(\frac{n\pi y}{b}\right) \quad (m, n \ge 1)$$

2. **Modo TE:** $E_z = 0$, hallar $B_z$. Condición de Neumann: $\frac{\partial B_z}{\partial n}|_{\partial\Omega} = 0$.

$$B_z(x, y) = B_0 \cos\left(\frac{m\pi x}{a}\right)\cos\left(\frac{n\pi y}{b}\right) \quad (m, n \ge 0, m{+}n > 0)$$

3. **Corte:** Frecuencia de corte límite $\omega_{mn} = c\sqrt{(\frac{m\pi}{a})^2 + (\frac{n\pi}{b})^2}$.
4. **Transversales:** Hallar $\mathbf{E}_\perp, \mathbf{B}_\perp$ derivando componentes longitudinales:

$$\mathbf{E}_\perp = \frac{1}{\gamma^2}[ik\nabla_\perp E_z - i\omega(\hat{z}\times\nabla_\perp B_z)]$$

## 2. Transformación de Campos ($F^{\mu\nu}$)

Métrica $\eta_{\mu\nu} = \text{diag}(1, -1, -1, -1)$. Transformación tensorial: $F' = \Lambda F \Lambda^T$.

1. **Tensor $F^{\mu\nu}$:** $F^{0i} = -E_i/c$, $F^{ij} = -\epsilon_{ijk}B_k$.
2. **Boost (eje $x$):** $\Lambda^0{}_0 = \gamma, \Lambda^0{}_1 = \Lambda^1{}_0 = -\beta\gamma, \Lambda^1{}_1 = \gamma$.
3. **Campos Rotados / Transformados:**

$$E'_x = E_x, \quad E'_y = \gamma(E_y - vB_z), \quad E'_z = \gamma(E_z + vB_y)$$

$$B'_x = B_x, \quad B'_y = \gamma(B_y + \frac{v}{c^2}E_z), \quad B'_z = \gamma(B_z - \frac{v}{c^2}E_y)$$

4. **Invariantes:** Evaluar $E^2 - c^2B^2$ y $\mathbf{E}\cdot\mathbf{B}$ para chequear consistencia.

## 3. Radiación de Dipolo Eléctrico Oscilante

Fuente armónica $\rho(\mathbf{r}, t) = \rho_0(\mathbf{r})e^{-i\omega t}$. Zona lejana ($r \gg \lambda \gg d$).

1. **Momento Dipolar:** $\mathbf{p} = \int \mathbf{r}\rho_0(\mathbf{r})d^3r$.
2. **Campos de Radiación:** $\mathbf{B} = \frac{\mu_0\omega^2}{4\pi c}(\hat{r}\times\mathbf{p})\frac{e^{ikr}}{r}$ y $\mathbf{E} = c(\mathbf{B}\times\hat{r})$.
3. **Vector de Poynting:** $\langle\mathbf{S}\rangle = \frac{\mu_0\omega^4|p|^2}{32\pi^2 c}\frac{\sin^2\theta}{r^2}\hat{r}$ (si $\mathbf{p} = p\hat{z}$).
4. **Potencia Total:** Integral de flujo angular $\implies P = \frac{\mu_0\omega^4|p|^2}{12\pi c}$.

## 4. Relaciones de Kramers-Kronig

Consecuencia directa de la causalidad lineal ($\chi(t) = 0$ para $t < 0$).

$$\text{Re } \epsilon(\omega) - 1 = \frac{2}{\pi}\mathcal{P}\int_0^\infty \frac{\Omega \text{Im } \epsilon(\Omega)}{\Omega^2 - \omega^2}d\Omega$$